Heqin Zhu, PhD

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Research Interests

My research focuses on AI for Biology, developing multimodal foundation models and self-improving AI systems to understand biomolecular structures, functions, and interactions, and to enable molecular design and AI-driven drug discovery, with the following research topics:

- **Multimodal Biological Foundation Model**: Structure-guided RNA foundation model: structRFM [R8] (Nature).
- **Molecular Design and Drug Discovery**: RNA and DNA aptamer design: NACraft [R10].
- **Molecular Interaction and Scoring**: RNA–compound interaction prediction and binding-site localization: ReTIF [R11] (bioRxiv).
- **Structure and Function Prediction**: RNA secondary structure prediction: BPfold [R6] (Nature Communications), NCfold [R9] (ICLR 2026); IRES identification: IRESeek [R7].
- **Recursive Self-Improvement**: Self-evolving scientific agent for sustained research across heterogeneous tasks: ADMET-EvO [C9] (arXiv).

Previously, I developed universal models for **Medical Image Analysis**: domain adaptation: GU2Net [R1, R2] (MICCAI 2021, BMEF), DATR [R4]; few-shot learning: UOD [R5] (MICCAI 2023).

Education and Employment

The Chinese University of Hong Kong (CUHK)
Postdoc, Department of Computer Science and Engineering (CSE) 11/2026 – Present
- Advisor: Prof. Pheng Ann Heng

University of Science and Technology of China (USTC)
Ph.D., Biomedical Engineering 09/2023 – 06/2026
- Advisor: Prof. Shaohua Kevin Zhou

Institute of Computing Technology (ICT), Chinese Academy of Sciences (CAS) & University of Chinese Academy of Sciences (UCAS)
M.S., Computer Science and Technology 09/2020 – 06/2023
- Advisor: Prof. Shaohua Kevin Zhou

University of Science and Technology of China (USTC)
B.S., Computer Science and Technology 09/2016 – 06/2020
- Hua Xia Talent Program in Computer Science and Technology

Selected Publications

R=Representative, C=Collaborative

# denotes co-first author and * denotes co-corresponding author. For full list, please refer to Google Scholar.

Representative Papers

[R11] Ao Chang#, **Heqin Zhu**#, Haobin Chen, Chenxi Wang, Xin Wang, Xiaoyu Zeng, Yilin Ding, Peng Xiong*, and S. Kevin Zhou*. "ReTIF: Granularity-Aware Multitask Interaction Routing for RNA-Compound Interaction Prediction and Binding-Site Localization." bioRxiv (2026). (Preprint). [bioRxiv]

[R10] **Heqin Zhu**, Jiaqi Wang, Weibo Zhao, YuZhi Xu, Huang Su, Jianmin Wang, Qinghan Wang, Yuntao Yu, Ziyi You, Gang Du, Pheng Ann Heng*, Liqin Zhang*, and Odin Zhang*. "NACraft: Programmatic nucleic-acid aptamer design via all-atom structure-model feedback" (Preprint). [bioRxiv; Code]

[R9] **Heqin Zhu**#, Ruifeng Li#, Ao Chang, Mingqian Li, Hongyang Chen*, Peng Xiong*, and S. Kevin Zhou*. "Toward Accurate RNA Non-Canonical Structure Prediction: The NC-Bench Benchmark and the NCfold Framework." (ICLR 2026). [bioRxiv; Code]

[R8] **Heqin Zhu**, Ruifeng Li, Ao Chang, Haobin Chen, Feng Zhang, Fenghe Tang, Tong Ye, Xin Li, Yunjie Gu, Peng Xiong*, and S. Kevin Zhou*. "An open structure-guided RNA foundation model for generalizable prediction." (Nature). [Nature; bioRxiv; Code]

[R7] Feng Zhang#, **Heqin Zhu**#, Jiayin Gao, Jie Hu, Ke Chen, Shaohua Kevin Zhou*, and Peng Xiong*. "IRESeek: structure-informed deep learning method for accurate identification of internal ribosome entry sites in circular RNAs." NAR Genomics and Bioinformatics 7, no. 4 (2025): lqaf210. (NAR Genomics and Bioinformatics). [Paper;Code]

[R6] **Heqin Zhu**, Fenghe Tang, Quan Quan, Ke Chen, Peng Xiong*, and S. Kevin Zhou*. "Deep generalizable prediction of RNA secondary structure via base pair motif energy." Nature Communications 16, (2025): 5856. (Nat. Commun., 2025). [Paper; Code]

[R5] **Heqin Zhu**, Quan Quan, Qingsong Yao, Zaiyi Liu, and S. Kevin Zhou. "Uod: Universal one-shot detection of anatomical landmarks." In International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 24-34. Cham: Springer Nature Switzerland, 2023. (MICCAI 2023). [Paper; Code]

[R4] **Heqin Zhu**, Qingsong Yao, and S. Kevin Zhou. "Datr: Domain-adaptive transformer for multi-domain landmark detection." arxiv preprint arxiv:2203.06433 (2022). (Preprint). [Paper; Code]

[R3] **Heqin Zhu**, Xu Sun, Yuexiang Li, Kai Ma, S. Kevin Zhou*, and Yefeng Zheng*. "DFTR: Depth-supervised fusion transformer for salient object detection." arxiv preprint arxiv:2203.06429 (2022). (Preprint). [Paper; Code]

[R2] **Heqin Zhu**, Qingsong Yao, Li Xiao, and S. Kevin Zhou. "Learning to Localize Cross-Anatomy Landmarks in X-Ray Images with a Universal Model." BME Frontiers 2022 (2022): 9765095. (BMEF 2022). [Paper; Code]

[R1] **Heqin Zhu**, Qingsong Yao, Li xiao, and S. Kevin Zhou. "You only learn once: Universal anatomical landmark detection." In Medical Image Computing and Computer Assisted Intervention, pp. 85-95. Springer International Publishing, 2021. (MICCAI 2021). [Paper; Code]

Collaborative Papers

[C9] Yiling Zhou#, Yilin Wang#, Jianmin Wang, **Heqin Zhu**, Zirui Wang, Chang-yu Hiesh, Kejun Ying, Jiaqi Wang*, Yuzhi Xu*, Tingjun Hou*, and Odin Zhang*. "ADMET-EvO: a self-evolving scientific agent for sustained research across heterogeneous tasks." arXiv preprint arXiv:2609.10121 (2026). (Preprint). [Paper]

[C8] Fenghe Tang, Chengqi Dong, Wenxin Ma, Zikang Xu, **Heqin Zhu**, Zihang Jiang, Rongsheng Wang, Yuhao Wang, Chenxu Wu, and Shaohua Kevin Zhou. "U-Bench: A Comprehensive Understanding of U-Net through 100-Variant Benchmarking." In Advances in Neural Information Processing Systems (NeurIPS), 2026. (NeurIPS 2026). [Paper]

[C7] Quan Quan#, Qingsong Yao#, **Heqin Zhu**, and S. Kevin Zhou. "IGU-Aug: Information-guided unsupervised augmentation and pixel-wise contrastive learning for medical image analysis." IEEE Transactions on Medical Imaging (2024). (TMI 2024).

[C6] Quan Quan#, Qingsong Yao#, **Heqin Zhu**, Qiyuan Wang, and S. Kevin Zhou. "Which images to label for few-shot medical image analysis?." Medical Image Analysis 96 (2024): 103200. (MIA 2024).

[C5] Huang Zhen#, Han Li#, Shitong Shao, **Heqin Zhu**, Huijie Hu, Zhiwei Cheng, Jianji Wang, and S. Kevin Zhou. "PELE scores: pelvic X-ray landmark detection with pelvis extraction and enhancement." International Journal of Computer Assisted Radiology and Surgery 19, no. 5 (2024): 939-950. (IJCARS 2024).

[C4] Xinyi Wang, Zikang Xu, **Heqin Zhu**, Qingsong Yao, Yiyong Sun, and S. Kevin Zhou. "SIX-Net: Spatial-Context Information miX-up for Electrode Landmark Detection." In International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 338-348. Cham: Springer Nature Switzerland, 2024. (MICCAI 2024).

[C3] Fenghe Tang, Ronghao Xu, Qingsong Yao, Xueming Fu, Quan Quan, **Heqin Zhu**, Zaiyi Liu, and S. Kevin Zhou. "Hyspark: Hybrid sparse masking for large scale medical image pre-training." In International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 330-340. Cham: Springer Nature Switzerland, 2024. (MICCAI 2024).

[C2] Pengbo Liu, Hu Han, Yuanqi Du, **Heqin Zhu**, Yinhao Li, Feng Gu et al. "Deep learning to segment pelvic bones: large-scale CT datasets and baseline models." International Journal of Computer Assisted Radiology and Surgery 16 (2021): 749-756. (IJCARS 2021).

[C1] Yuanyuan Lyu, Haofu Liao, **Heqin Zhu**, and S. Kevin Zhou. "A 3 DSegNet: anatomy-aware artifact disentanglement and segmentation network for unpaired segmentation, artifact reduction, and modality translation." In International Conference on Information Processing in Medical Imaging, pp. 360-372. Cham: Springer International Publishing, 2021. (IPMI 2021).

Selected Research Projects

NACraft: Structure-Guided Nucleic-Acid Aptamer Design [R10] 08/2026

- **Lead developer**; built a training-free framework for de novo, similarity-guided, and target-selective RNA/DNA aptamer design by optimizing structure-model feedback and refining sequences with NA-MPNN.
- Reached maximum AF3 ipTM values of **0.63–0.88** across five protein targets; **69.44%** of paired candidates favoured EGFR over HER2. Outperformed ODesign on **10/11** NA-12 targets and **17/20** therapeutic target–length settings.

structRFM: Structure-Guided RNA Foundation Model [R8] Nature

- **First author**; led data construction, model development, evaluation, and open-source release. Pre-trained on **21 million RNA sequence–structure pairs** using multi-source structural annotations (MUSES) and structure-guided masked language modeling (SgMLM).
- Achieved state-of-the-art results on multiple RNA structure and function benchmarks and ranked among the **top four** in zero-shot homology classification.

BPfold: RNA Structure Prediction with Thermodynamic Priors [R6] Nature Communications 07/2025

- **First author**; led method development and reproducible software release, combining base-pair motif energy priors, base-pair attention, and physically constrained structure refinement.
- Improved F1 and INF by **approximately 5%** over SPOT-RNA on bpRNA-TS0; achieved the best **F1 of 0.689** on **10,791 RNAs from unseen families** in Rfam12.3–14.10.

Open-source Software and Resources

NACraft GitHub — De novo, similarity-guided, and target-selective RNA/DNA aptamer design.

structRFM GitHub — Open foundation model for RNA structure and function prediction.

BPfold GitHub — Generalizable RNA secondary structure prediction with thermodynamic priors.

USTC Course Resources GitHub — **Lead and maintainer** (2019 – Present); curated course notes, exams, code, and learning resources; 16.2K GitHub stars.

Honors and Awards

President's Excellent Award, Chinese Academy of Sciences	2026
Provincial Outstanding Graduate (Ph.D., Class of 2026), Anhui Province	2026
Outstanding Graduate (Ph.D., Class of 2026), USTC	2026
Outstanding Academic Presentation Award (1st of 10), The 7th USTC Academic Forum for Engineering Graduate Students	2026
National Scholarship for Doctoral Students, Ministry of Education of China	2025
Second Prize for Oral Presentation, The 3rd National Conference on Biomolecular Structure Prediction and Simulation	2025
Suzhou Industrial Park Scholarship, USTC	2025
Merit Student Award, UCAS & ICT	2023
Graduate Academic Scholarship, UCAS & ICT and USTC	2020 – 2025
Outstanding Student Award, USTC	2018 – 2019
Institute of Chemistry Talent Award, USTC	2017

Internship Experience

Tencent, JARVIS Lab, Research Intern Shenzhen, China 07/2021 – 11/2021

- Developed a depth-supervised feature-fusion transformer for salient object detection [R3].

Talks and Presentations

MSD R&D China Innovation Collaboration Center (MCICC) 06/15/2026 Virtual

Multimodal AI for RNA modeling and foundation model PDF

The 7th USTC Academic Forum for Engineering Graduate Students & the "Dechuang · Jingjian Innovation" Doctoral Academic Forum 06/13/2026 Hefei, China

Structure-guided RNA foundation model for structure and function prediction PDF

Outstanding Academic Presentation Award (1st of 10)

USTC Course on Digital Healthcare Technology and Applications 12/2025
Suzhou, China

Multimodal AI for RNA modeling and foundation model PDF

The 3rd National Conference on Biomolecular Structure Prediction and Simulation 05/2025
Changchun, China

Deep generalizable prediction of RNA secondary structure via base pair motif energy PDF
Outstanding Academic Presentation Award (2nd of 12)

Teaching and Mentoring

Teaching

Teaching Assistant: Biomolecular Structure Modeling, USTC 2024
Teaching Assistant: Open Practice in Electronic Information, USTC 2023

Research Mentoring

Ao Chang (Ph.D. student): Mamba-based RNA language pre-training 2025 – 2026
Haobin Chen (M.S. student): Agents for lncRNA literature retrieval 2025 – 2026
Duofan Tu (M.S. student): Medical agents for clinical diagnosis 2024 – 2026
Feng Zhang (M.S. student): IRES identification in circular RNAs (IRESeek) 2024 – 2026
Xiaoqian Zhou (M.S. student): Hybrid-attention anatomical landmark detection 2022 – 2024
Zhen Huang (M.S. student): Pelvic X-ray landmark detection (PELE scores) 2021 – 2023

Academic Service

- **Academic Reviewer**: NeurIPS, ICLR, MICCAI, TCSVT
- **Volunteer Activities**
  - Medical Augmented Reality Summer School (Suzhou, 2024)
  - Dushu Lake Symposium on Medical Image Computing (Suzhou, 2023)

Technical Skills

Methods: Foundation models, multimodal learning, large language models, diffusion models

Programming & Tools: Python, PyTorch, C++, Git, LaTeX, Matplotlib

References

Shaohua Kevin Zhou — PhD Advisor skevinzhou@ustc.edu.cn
Professor, University of Science and Technology of China

Peng Xiong — PhD Co-Advisor xiongxp@ustc.edu.cn
Professor, University of Science and Technology of China

Pheng Ann Heng — Postdoc Advisor pheng@cse.cuhk.edu.hk
Professor, The Chinese University of Hong Kong

Wanxiang Shen — Collaborator shenwx25@zju.edu.cn
Professor, Zhejiang University